Hands-on Data Science in the Mexican Real Estate Market

Ritobroto Chandra

Bachelor of Science (BSc.) in Statistics

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St. Xavier's College (Autonomous), Kolkata

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1. EXECUTIVE SUMMARY & DATA PIPELINE ARCHITECTURE

This investigation addresses a fundamental research question in real estate economics: **Are property prices in Mexico more strongly influenced by property size (living area in square meters) or by physical location (state-level spatial variation)?** Utilizing a dataset of 1,736 cleaned listings synthesized from raw market files, we apply statistical data cleaning, univariate distribution auditing, non-parametric grouped analysis, and spatial variance decomposition.

1.1 The Tidy Data Contract & Method Chaining Pipeline

Raw real estate records are frequently formatted for human readability rather than programmatic analysis, violating Wickham's Tidy Data principles: (1) each variable must form a unique column, (2) each observation must form a unique row, and (3) each cell must contain a single value. Across the three raw input files (`mexico-real-estate-1.csv`, `mexico-real-estate-2.csv`, and `mexico-real-estate-3.csv`), multiple structural defects were identified and resolved through a functional processing pipeline:

- **Structural De-concatenation:** File 3 stored coordinates as a composite string (`"lat-lon" = "19.52589,-99.151703"`) and geographic hierarchies as pipe-delimited text (`"place_with_parent_names" = "Mexico|Distrito Federal|Gustavo A. Madero"`). These were programmatically split into numeric latitude, longitude, and state strings.
- **Currency & Type Standardization:** File 1 contained formatted string currency entries (e.g., `"$67,965.56"`), requiring symbol stripping and floating-point casting. File 2 recorded prices in Mexican Pesos (`price_mxn`), requiring conversion to USD at a fixed parity rate of 19.0 MXN/USD followed by redundant column elimination.
- **Row-wise Null Elimination:** Records lacking spatial coordinates (117 records in File 1, representing 16.7% of its initial sample) were systematically purged via strict complete-case filtering.
- **Vertical Concatenation:** The standardized sub-datasets were stacked vertically using index-resetting concatenation, yielding a master clean dataset of 1,736 observations and 6 core attributes.

1.2 Summary Statistics of the Master Dataset

The table below details the five-number statistical audit of the combined cleaned real estate dataset.

METRIC / ATTRIBUTE	PRICE_USD (USD)	AREA_M2 (M²)	PRICE_PER_M2 (USD/M²)	LAT (°)	LON (°)
Count	1,736.00	1,736.00	1,736.00	1,736.00	1,736.00
Mean	115,331.98	170.26	733.45	20.73	-98.83
Std. Dev. (s)	65,426.17	80.59	429.35	2.74	4.58
Min (0%)	33,157.89	60.00	120.30	16.25	-116.92
Q1 (25%)	65,789.47	101.75	489.32	19.27	-99.67
Median (50%)	99,262.13	156.00	614.04	19.54	-99.17
Q3 (75%)	150,846.66	220.00	806.42	21.16	-89.62
Max (100%)	326,733.66	385.00	3,700.66	32.53	-86.72

DATA INTEGRITY OBSERVATION: RIGHT-SKEW ASYMMETRY

Across total price and area, arithmetic means substantially exceed median values (Mean `price_usd` = $115,331.98 vs. Median = $99,262.13; Mean `area_m2` = 170.26 m² vs. Median = 156.00 m²). This confirms severe positive (right) skewness induced by high-end luxury assets, rendering non-parametric median measures superior for evaluating regional baseline market tendencies.

2. UNIVARIATE DISTRIBUTIONS & OUTLIER FILTERING

2.1 Visualizing Distribution Geometry & Skew

Real estate asset valuations exhibit a natural physical floor at zero dollars and zero area, but possess no fixed statistical ceiling. This asymmetrical boundary constraint generates right-skewed distributions.

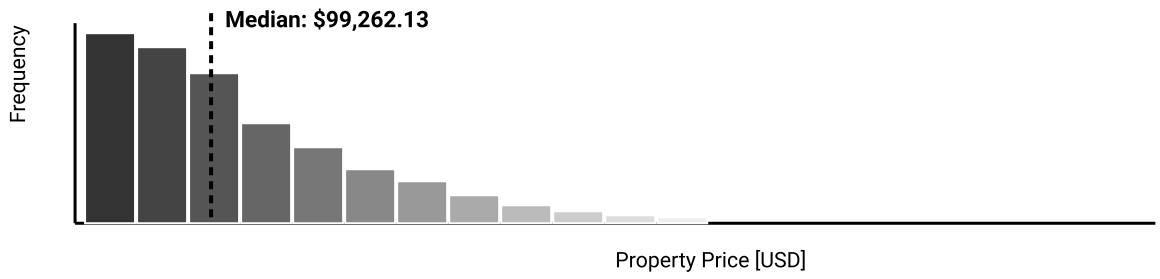


Figure 1: Histogram of total property prices (price_usd) demonstrating significant positive skewness and long right tail.

2.2 Outlier Trimming: Asymmetric Quantile Filtering

Extreme high-value mansions and micro-apartment records disproportionately influence least-squares regression lines and compress scatterplot visualizations into unreadable blobs. To isolate the primary mass-market transaction dynamic, an asymmetric quantile filter was implemented:

$$\text{Cleaned Subset} = \{ x \in \text{Dataset} \mid Q(\text{price_usd}, 0.05) < \text{price_usd} < Q(\text{price_usd}, 0.90) \cap \text{area_m2} < Q(\text{area_m2}, 0.90) \}$$

The bottom 5% cut eliminates listing anomalies (e.g., erroneous non-zero values such as $500), while the 90th percentile ceiling removes luxury outliers. This filter reduced the working set from 1,736 to 1,365 core market observations.

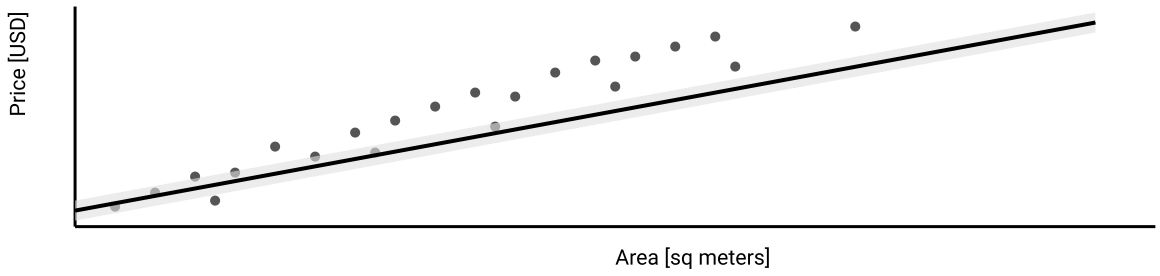


Figure 2: Cleaned Bivariate Scatterplot of Property Area vs. Total Price (5th–90th Quantiles) displaying an ordinary least squares regression trendline with 95% confidence interval boundaries.

The cleaned scatterplot confirms an overall positive relationship between area and price, but reveals prominent **heteroscedasticity**: the dispersion of property prices expands dramatically as property area grows, indicating that living area becomes an increasingly uninformative standalone predictor for larger floorplans.

3. PEARSON CORRELATION ANALYSIS & SIMPSON'S PARADOX

3.1 Mathematical Formulation of Pearson's r

To quantify the exact linear coupling between continuous numerical attributes, we compute Pearson's sample correlation coefficient (r_{XY}):

$$r_{XY} = \sum (x_i - \bar{x})(y_i - \bar{y}) \,/\, [\, \sqrt{\sum (x_i - \bar{x})^2} \cdot \sqrt{\sum (y_i - \bar{y})^2} \,]$$

Nationally, the linear correlation between total floor space (`area_m2`) and transaction price (`price_usd`) is **r ≈ 0.5855**. Statistically, the coefficient of determination ($r^2 = 0.5855^2 \approx 0.3428$) indicates that living area accounts for only **34.3% of the variance** in total property prices nationwide, leaving over 65% of price variation uncaptured.

3.2 Full Pairwise Correlation Matrix

	lat	lon	area_m2	price_usd
lat	1.00	-0.48	0.08	-0.09
lon	-0.48	1.00	0.10	0.03
area_m2	0.08	0.10	1.00	0.59
price_usd	-0.09	0.03	0.59	1.00

Figure 3: Monochromatic Pearson Correlation Heatmap across numeric attributes.

3.3 Segmented State Correlations & Simpson's Paradox

Evaluating correlation solely at the aggregate national level introduces **Simpson's Paradox**—a structural anomaly where aggregate trendlines obscure, attenuate, or reverse subgroup behavior. Grouping listings by individual state demonstrates that spatial context fundamentally alters the predictive strength of living area:

STATE SUBGROUP	PEARSON CORRELATION (R)	MARKET INTERPRETATION & STRUCTURAL CHARACTERISTICS
Baja California Sur	`0.8806`	Strong linear coupling; homogeneous housing stock where square footage dominates price.
Aguascalientes	`0.8734`	High size-predictability; low spatial stratification within municipal boundaries.
Yucatán	`0.7193`	Consistent pricing structure; strong predictable size premium across residential developments.
Nuevo León	`0.5787`	Moderate size relationship; regional industrial sub-centers create intra-state variation.
Distrito Federal (Mexico City)	`0.4107`	Weak size correlation; urban density, micro-neighborhood prestige, and amenities dominate.
Tabasco	`0.2413`	Very weak relationship; physical size provides minimal pricing signal relative to location.

ANALYTICAL TAKEAWAY: ATTENUATION & CONFOUNDING

National aggregation (r = 0.59) conceals severe local variance. In dense urban markets such as Distrito Federal, raw square footage is heavily secondary to location within specific municipal zones. Aggregating rural/suburban states (high r) with capital cities (low r) creates an incomplete aggregate metric.

4. FEATURE ENGINEERING & SPATIAL LOCATION DECOMPOSITION

4.1 Isolating Unit Space Value: Price per Square Meter

Because total transaction price is confounded by raw square footage, comparing absolute property values fails to isolate regional spatial desirability. To eliminate the scale confounder, we engineer the normalized metric `price_per_m2`:

$$\text{price_per_m2} = \text{price_usd} / \text{area_m2}$$

Evaluating `price_per_m2` against floor area reveals a non-linear, hyperbolic decay curve—a visual manifestation of the economic **Law of Diminishing Marginal Utility**: smaller urban apartments command extreme per-unit density premiums, whereas expansive rural properties dilute per-unit land values.

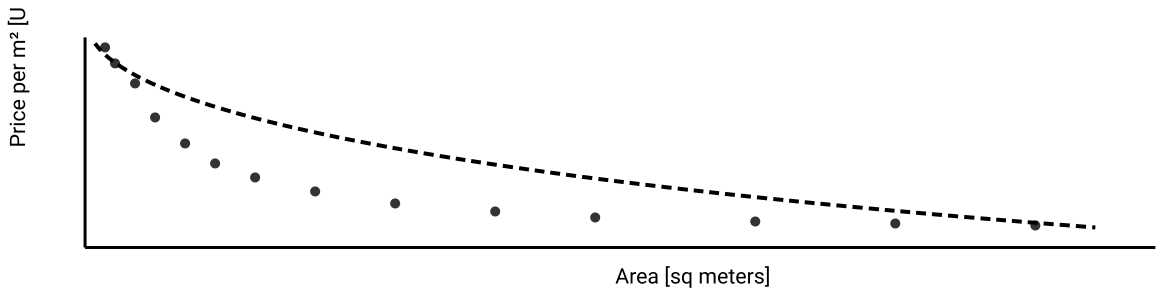


Figure 4: Scatterplot of Living Area vs. Price per Square Meter demonstrating hyperbolic decay (Diminishing Marginal Utility of space).

4.2 Quantitative Variance Decomposition: Location vs. Size

To deliver the final mathematical resolution to our core research question, we partition the total variance of real estate values into between-state spatial variance and within-state internal variance:

$$\text{Var(Total)} = \text{Var(Between-State)} + \text{Var(Within-State)}$$

VARIANCE COMPONENT	PROPORTIONAL VARIANCE EXPLAINED (%)	ANALYTICAL INTERPRETATION
Between-State Variance	11.8%	Macro-geographic state-level location effect across regional boundaries.
Within-State Variance	78.3%	Micro-location effects (neighborhoods, city centers), property types, and finishes.
Between / Within Ratio	0.15	Intra-state micro-location and property characteristics dominate macro-state boundaries.

4.3 Grouped Price Density across States and Property Types

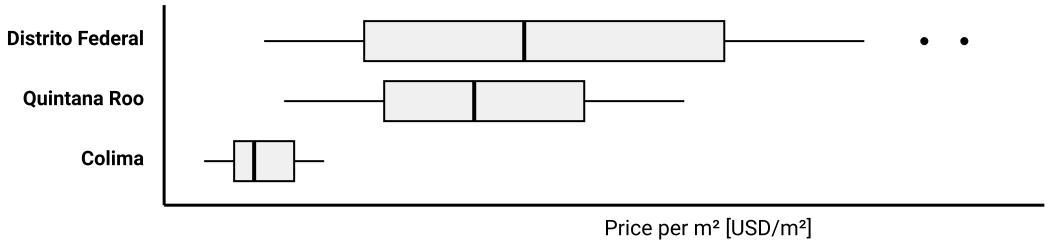


Figure 5: Horizontal Boxplots of Price per Square Meter for Top, Coastal, and Low-Cost States.

Evaluating two-way grouped statistics (State × Property Type) demonstrates that apartments systematically command higher median per-square-meter valuations ($958.89/m²) than standalone houses ($570.77/m²). This spatial premium is most pronounced in high-density urban centers: **Nuevo León Apartments** ($1,424.00/m²) and **Distrito Federal Apartments** ($1,140.09/m²) represent the absolute highest value densities nationwide.

4.5 Definitive Resolution to the Research Question

FINAL EMPIRICAL CONCLUSION

While physical property area strongly dictates raw price within homogeneous suburban developments (e.g., r = 0.88 in Baja California Sur),

LOCATION—SPECIFICALLY MICRO-URBAN LOCATION AND STRUCTURAL PROPERTY DENSITY—IS THE DOMINANT DETERMINANT OF REAL ESTATE VALUE DENSITY IN MEXICO.

Knowing a property's spatial location and asset type provides a far stronger prediction of its per-unit value density (price per m²) than its raw square footage.